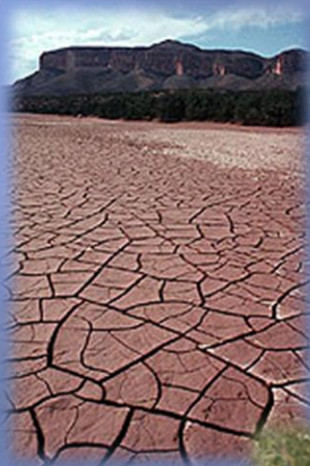
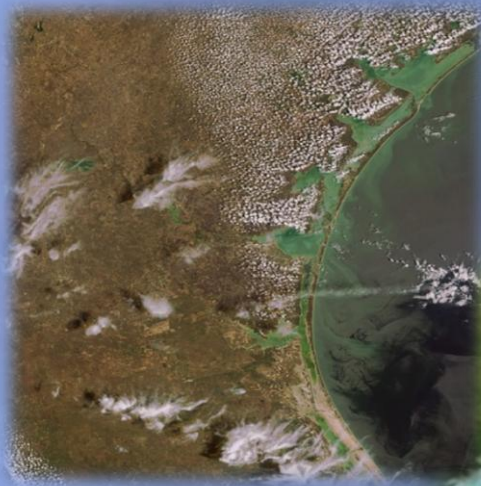


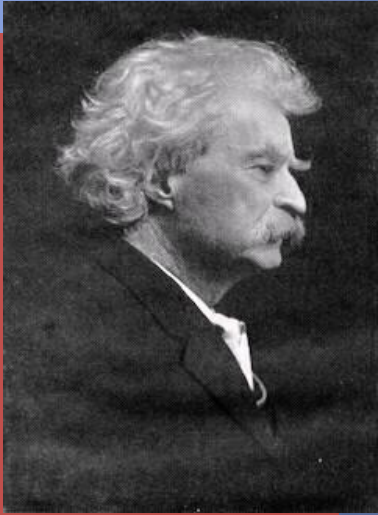
Adaptive Management and Gulf Restoration – A Perspective

Galveston Listening Session
June 2011



Dr. Larry McKinney
Director – Harte Research Institute
Texas A&M University Corpus Christi

Whiskey is for Drinking and Water is for Fighting

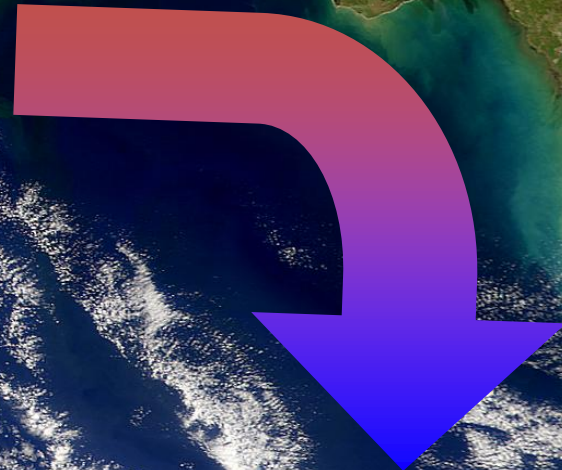


Mark Twain



**This is Not
A Texas Issue...**

**Water Issues
Are Mirror Images
On Either Side of
the Gulf**



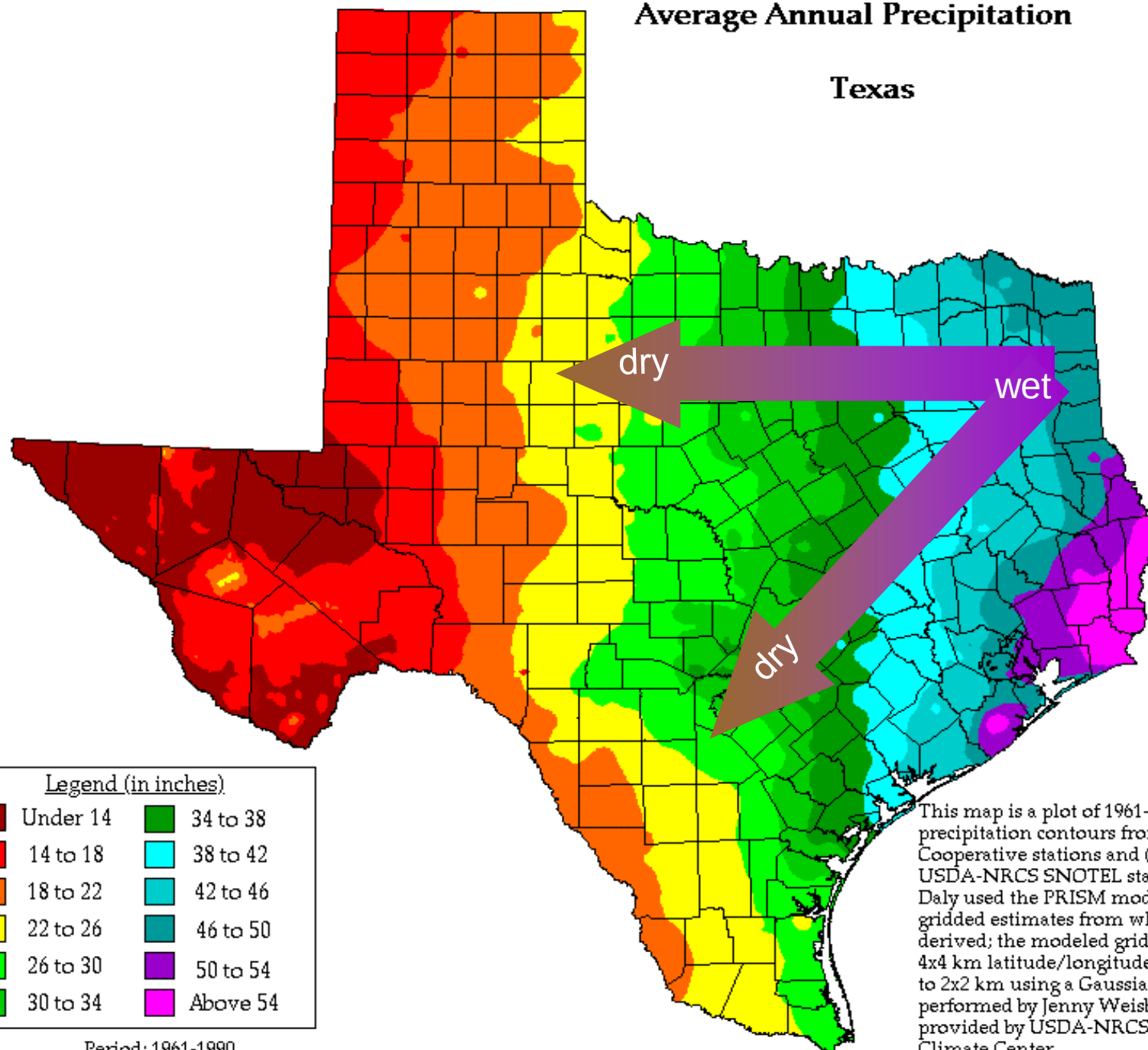
**Laguna
Madre**

**Mississippi
River**

**Florida
Bay**

Average Annual Precipitation

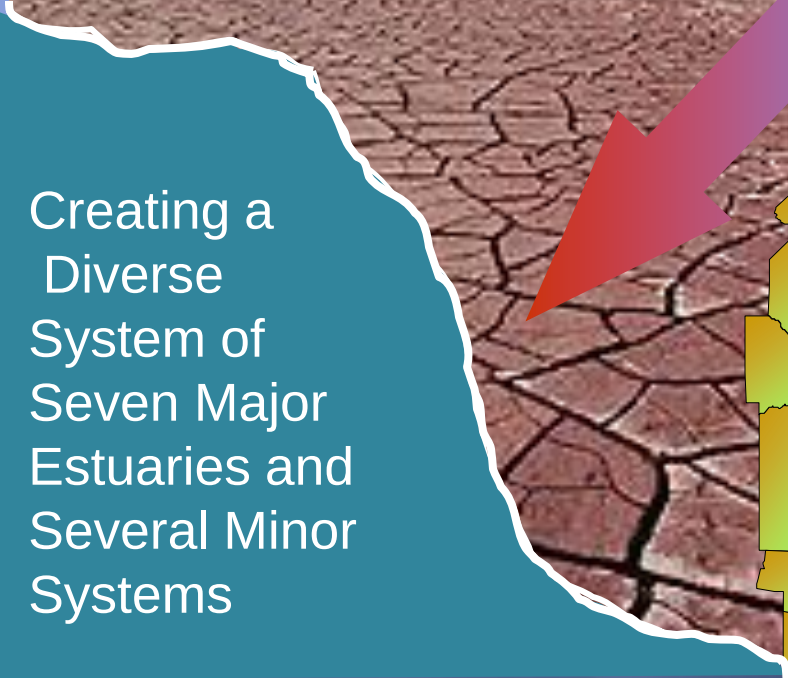
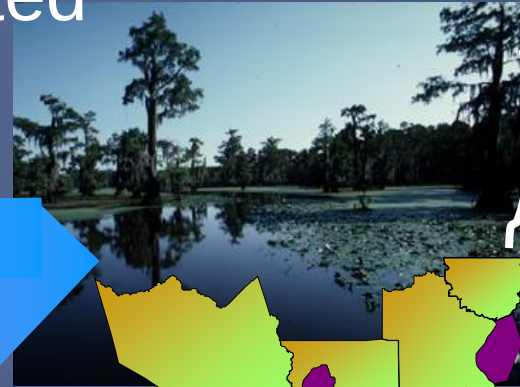
Texas



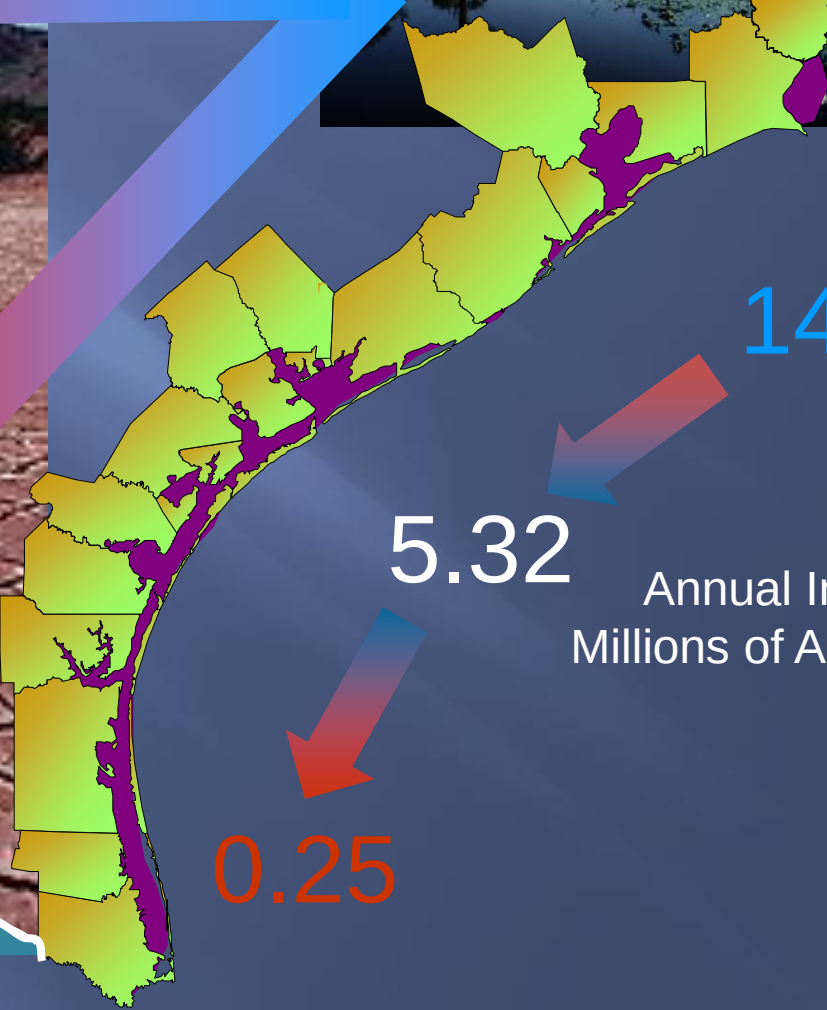
Period: 1961-1990

This map is a plot of 1961-1990 annual average precipitation contours from NOAA Cooperative stations and (where appropriate) USDA-NRCS SNOTEL stations. Christopher Daly used the PRISM model to generate the gridded estimates from which this map was derived; the modeled grid was approximately 4x4 km latitude/longitude, and was resampled to 2x2 km using a Gaussian filter. Mapping was performed by Jenny Weisburg. Funding was provided by USDA-NRCS National Water and Climate Center.

Texas Climate Gradient Is Reflected In Inflows to Texas Estuaries...



Creating a
Diverse
System of
Seven Major
Estuaries and
Several Minor
Systems



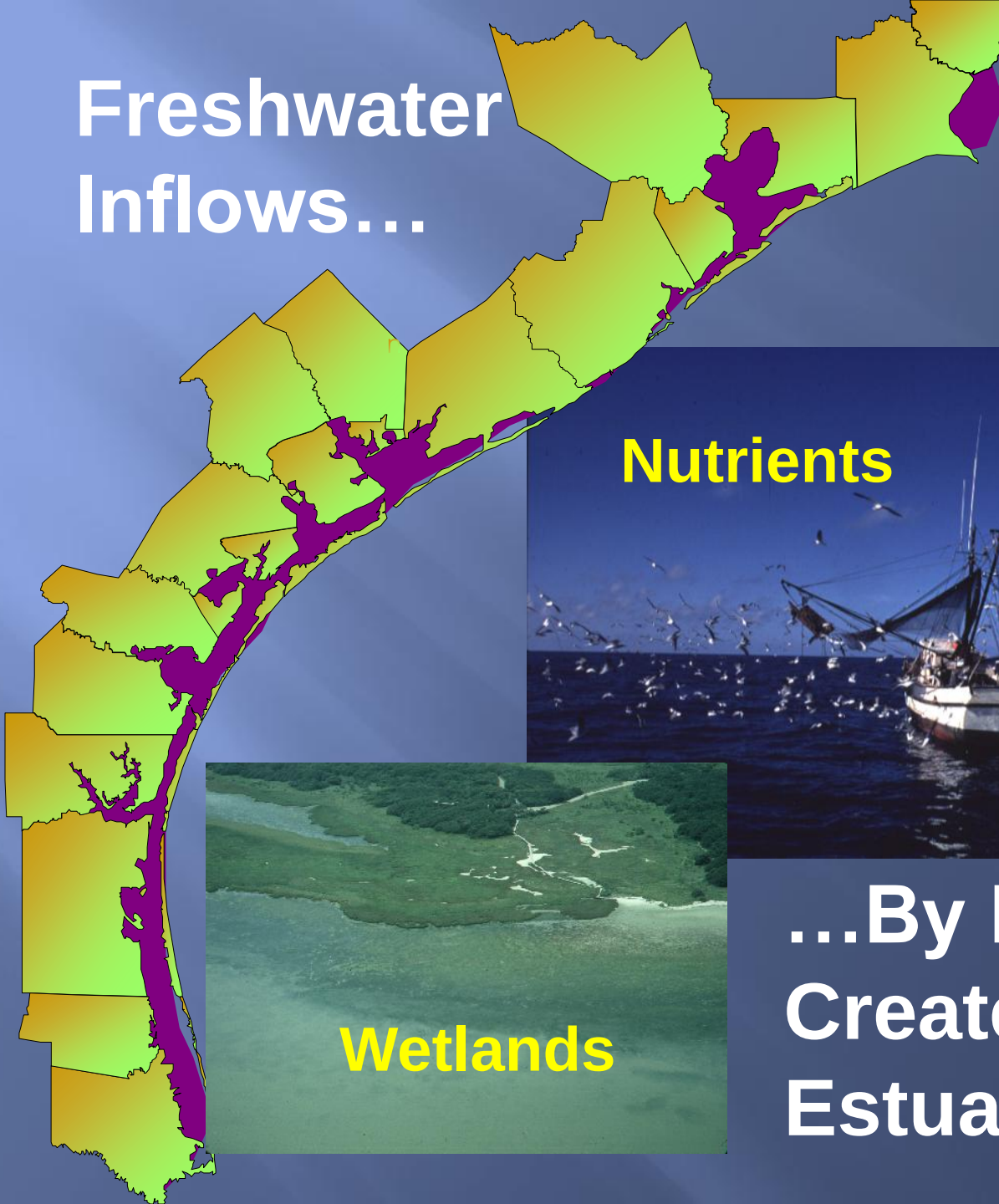
14.7

5.32

Annual Inflow
Millions of Acre Feet

0.25

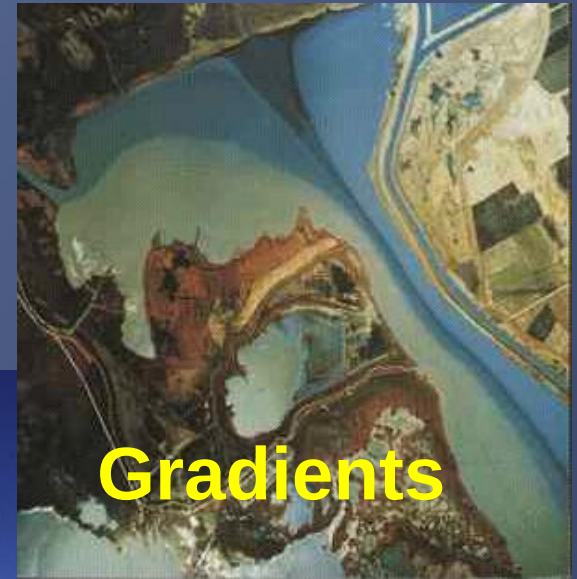
**Freshwater
Inflows...**



Nutrients



Gradients



Wetlands



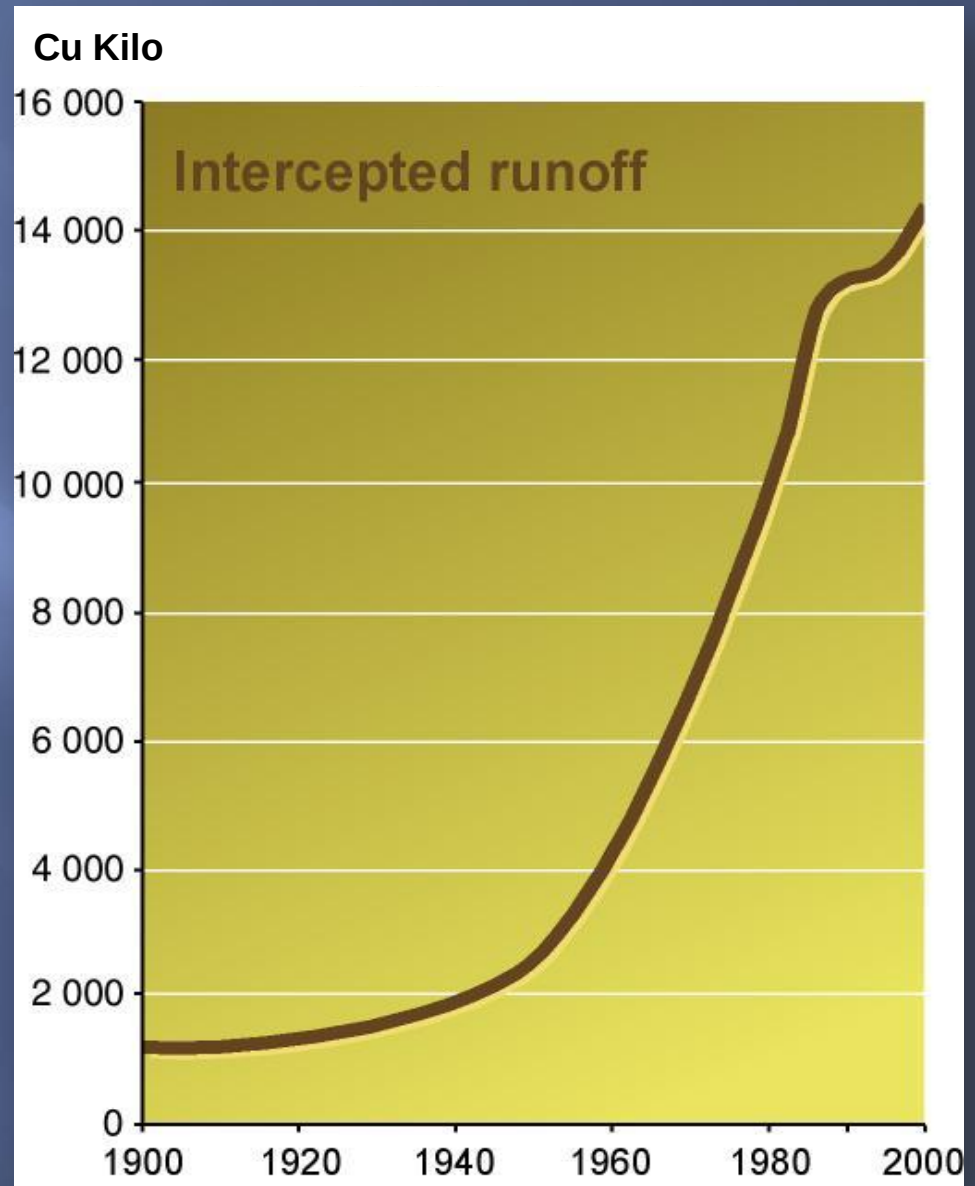
**...By Definition
Create and Sustain
Estuaries**



Competition For Water



- ▣ Amount of water in reservoirs quadrupled since 1960
- ▣ There is now 3-6 times as much water in reservoirs as in natural rivers
- ▣ Withdrawals from rivers and lakes doubled since 1960



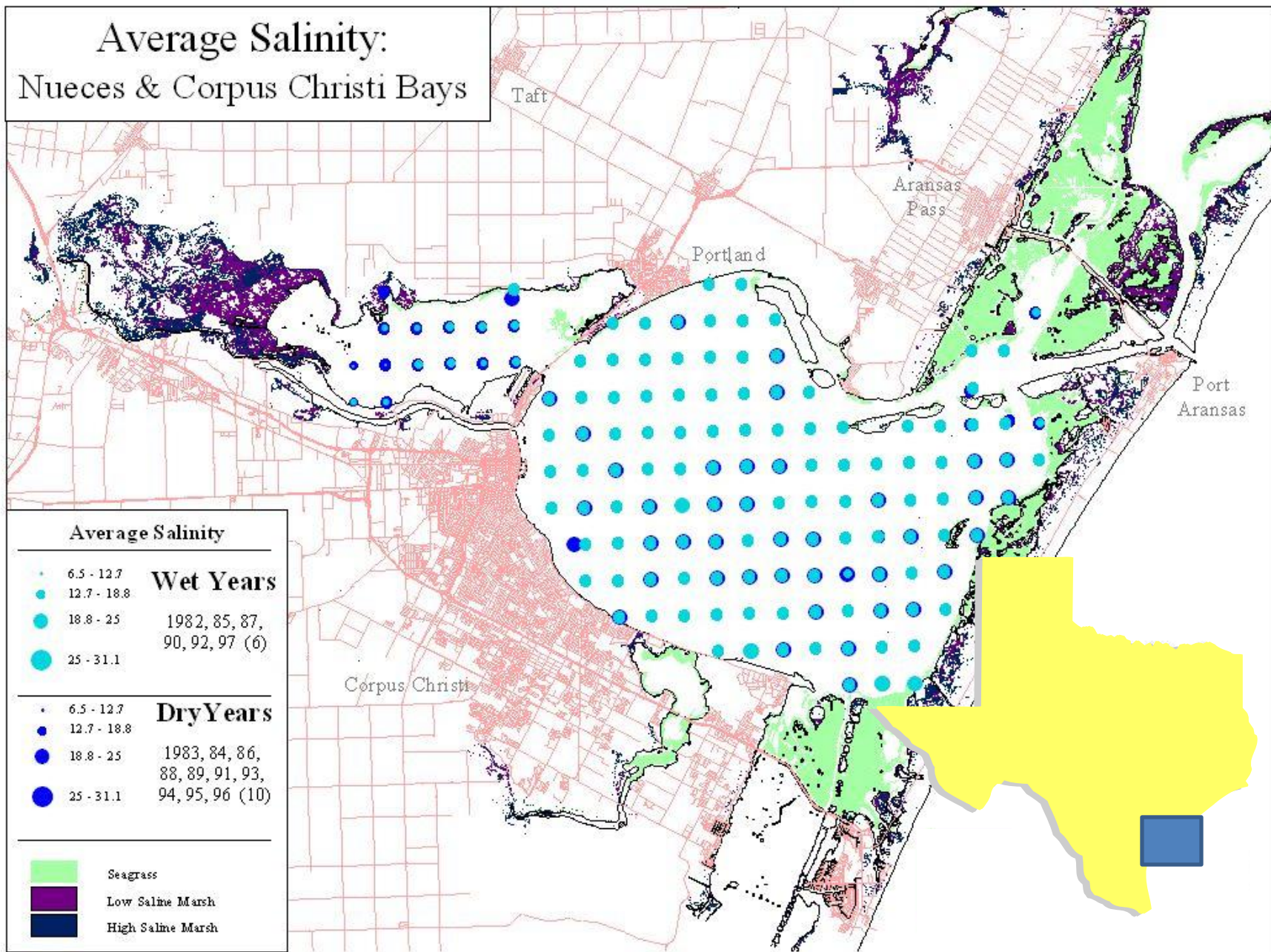
Source: <http://www.millenniumassessment.org/>



Texas Senate Bill 3 Basically put into Place an Adaptive Management Process to Secure Instream Flows and Freshwater Inflows to Protect Health and Productivity of Texas Rivers and Estuaries

- ▣ **Passed in 2007 by the 80th Regular Session of the Texas Legislature**
- ▣ **Created a basin-by-basin process for developing**
 - **Recommendations to meet instream flow needs for rivers**
 - **Recommendations to meet inflow needs for bays and estuaries**
 - **Environmental Flow Standards to be adopted by rule by the Texas Commission on Environmental Quality (TCEQ)**

Average Salinity: Nueces & Corpus Christi Bays



Adaptive management should be science driven but not science dominated. It is not possible to proceed with the certainty that many may demand. Such requests (for more certainty or more study) are often nothing more than delaying tactics. The primary purpose of adaptive management is to allow action in the face of a lack of certainty or scientific consensus, which is almost always the case in resource related issues. Once free of the illusion that science or technology can solve resource or conservation problems, a broad range of solutions become possible.

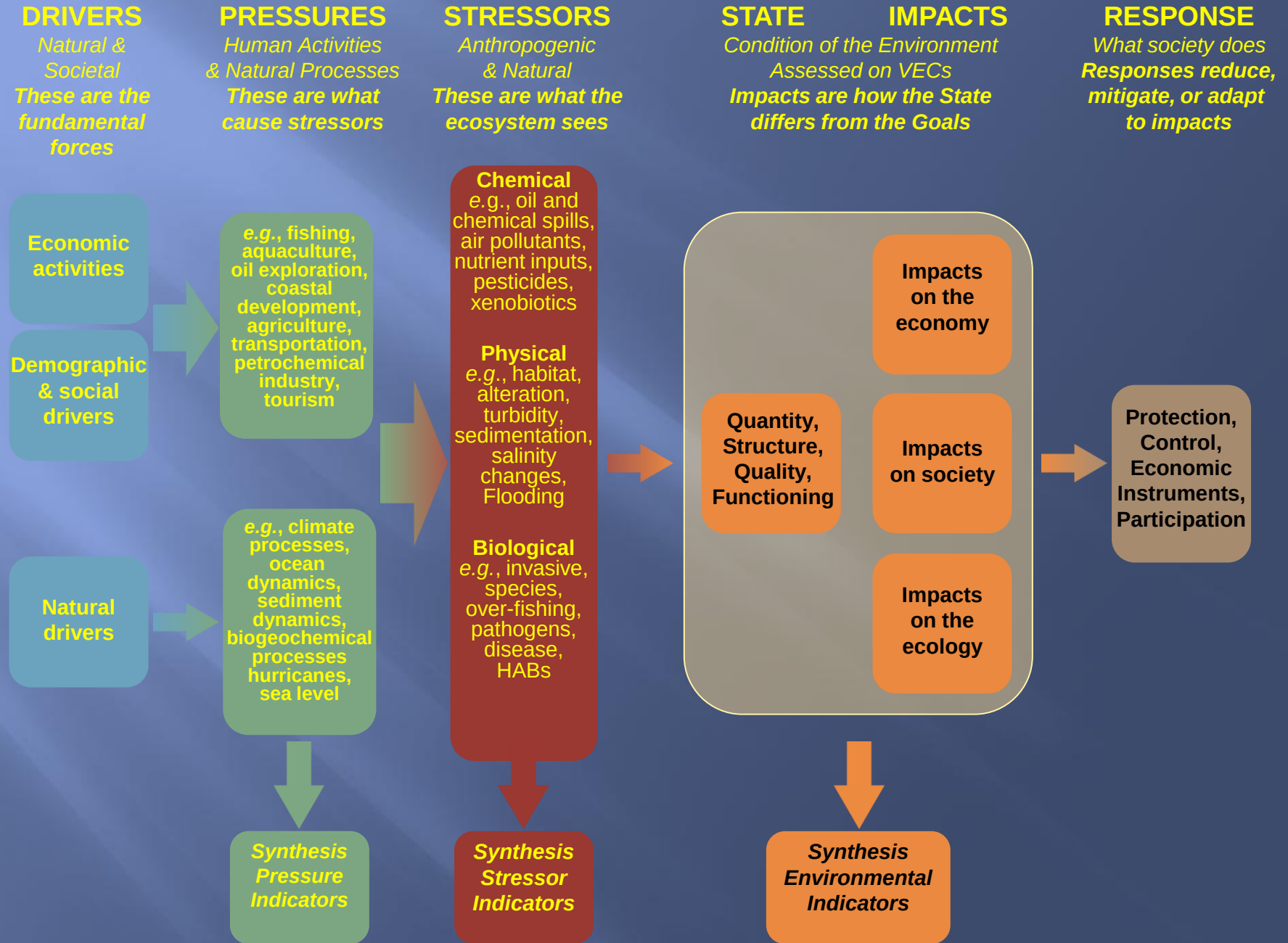
Rely on scientists to recognize problems but not to solve them. Resource issues are almost always multi-disciplinary in nature, science issues being relatively straightforward. It is the human element, socio-economic issues, that are at the root of the majority of problems impeding progress

Successful adaptive regulatory and adaptive management processes share several traits

Establish as specific a plan or process to be pursued, even in the face of uncertainties which will be raised, most likely by all stakeholders in the process. Never leave an open process that does not establish a *if not this, then what?*

A foundation to successful adaptive management is a means of monitoring the effects of actions taken under auspices of the process. You cannot know what management changes to propose unless you can objectively and accurately (to the degree possible) evaluate the effects of those actions taken.

GULF (DPSSIR) REPORT CARD - FRAMEWORK



Example Gulf of Mexico Report Card

(using DPSSIR framework & TNC/NGO Report)

DRIVERS	PRESSURES	STRESSORS	STATE	IMPACTS	RESPONSE
Natural & Anthropogenic <i>These are the fundamental forces</i>	Human Activities & Natural Processes <i>These are what cause stressors</i>	Anthropogenic & Natural <i>These are what the ecosystem sees</i>	Condition of the Environment Assessed on VECs <i>Impacts are how the State differs from the Goals</i>		What society does <i>Responses reduce, mitigate, or adapt to impacts</i>
Economic activities	Physical: - Coastal Development - Dredging - Shoreline Structures - Oil/Gas Exploration - Transportation	Physical: - Habitat Alteration - Changes in Hydrology - Harvesting - Changes in Salinity - Changes in Climate - Suspended Sediment - Noise	Fish & Wildlife - Fisheries Populations - Avian Populations - Marine Mammals - Terrapins & Turtles	Goal: Sustainable Fish & Wildlife - Species Diversity - Species Richness - Abundance - Productivity - Distribution	Restoration Goals: - Restore Water Quality - Ensure Freshwater Flows - Conserve Special Places - Incorporate Climate Change
Demographic & Social Drivers	Resource Extraction: - Commercial Fishing - Recreational Fishing Chemical Releases: - Petroleum spills - Chemical spills	Biological: - Introduced Species - Overfishing - Biological Competition - Disease - HAB	Habitats: - Wetlands - Mangroves - Oyster Reefs - Seagrasses - Coral Reefs - Barrier Islands - Intertidal Communities	Goal: Recover and Sustain Productive Habitats	Develop: - Environmental Report Card - Long-term Monitoring - Market-based Solutions Promote: - Citizen Stewardship - Environmental Outreach and Education
Natural Drivers	Natural Processes: - Loop Current - Hurricanes - Tropical Storms	Chemical: - Nutrients - Pesticides - Petroleum	Physical Features: Connectivity of Gulf with Coastal River Flows	Goal: Restore Physical Features	Implement: Adaptive Management

Example Gulf of Mexico Report Card

(using DPSSIR framework & TNC/NGO Report)

DRIVERS

Natural &
Anthropogenic

*These are the
fundamental forces*

**Economic
activities**

PRESSURES

Human Activities
& Natural Processes

*These are what
cause stressors*

Physical:

- Coastal Development
- Dredging
- Shoreline Structures
- Oil/Gas Exploration
- Transportation

STRESSORS

Anthropogenic
& Natural

*These are what
the ecosystem sees*

Physical:

- Habitat Alteration
- Changes in Hydrology
- Harvesting
- Changes in Salinity
- Changes in Climate
- Suspended Sediment
- Noise

Example Gulf of Mexico Report Card

(using DPSSIR framework & TNC/NGO Report)

STATE

Condition of the
Environment
Assessed on Valued
EOCSYSTEM
Components (VECs)

IMPACTS

*How the State
differs from the Goals*

RESPONSE

What society does
*Responses reduce,
mitigate, or adapt
to impacts*

Fish & Wildlife

- Fisheries Populations
- Avian Populations
- Marine Mammals
- Terrapins & Turtles

Goal: Sustainable Fish / Wildlife

- Species Diversity
- Species Richness
- Abundance
- Productivity
- Distribution

Restoration Goals:

- Restore Water Quality
- Ensure Freshwater Flows
- Conserve Special Places
- Incorporate Climate Change

Join us at...



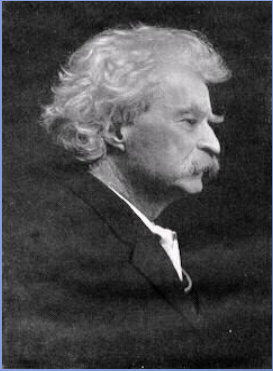
SUMMIT2011

STATE OF THE GULF OF MEXICO
HOUSTON TEXAS
DEC 4 – 8 2011



Charting a Course for the Future of America's Sea

Do the Right Thing...



...It Will Gratify Some People and Astound the Rest